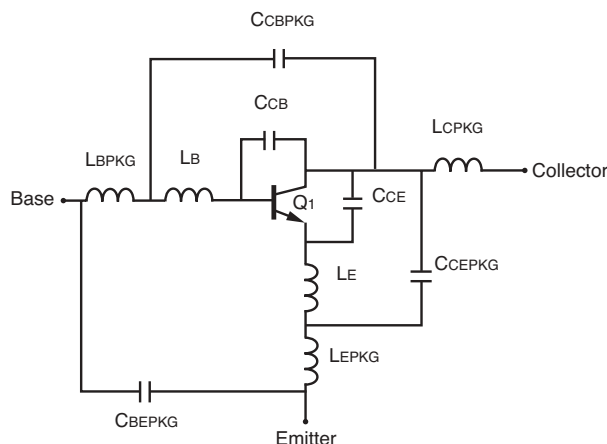


SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS⁽¹⁾

Parameters	Q1	Parameters	Q1
IS	4.429e-15	MJC	0.108
BF	331	XCJC	1
NF	1.141	CJS	0
VA	15	VJS	0.75
IKF	31e-3	MJS	0
ISE	5.324e-15	FC	0.8
NE	1.609	TF	4e-12
BR	17.10	XTF	10
NR	1.102	VTF	5
VAR	2.70	ITF	0.5
IKR	26.09e-3	PTF	20
ISC	100e-18	TR	0
NC	1.197	EG	1.11
RE	1.6	XTB	1.3
RB	1.0	XTI	5.2
RBM	50e-3	KF*	0
IRB	1e-4	AF*	1
RC	5.0		
CJE	459.9e-15		
VJE	767.5e-3		
MJE	64.7e-3		
CJC	109.4e-15		
VJC	0.67		

(1) Gummel-Poon Model

Life Support Applications

These NEC products are not intended for use in life support devices, appliances, or systems where the malfunction of these products can reasonably be expected to result in personal injury. The customers of CEL using or selling these products for use in such applications do so at their own risk and agree to fully indemnify CEL for all damages resulting from such improper use or sale.

CEL California Eastern Laboratories, Your source for NEC RF, Microwave, Optoelectronic, and Fiber Optic Semiconductor Devices.
4590 Patrick Henry Drive • Santa Clara, CA 95054-1817 • (408) 988-3500 • FAX (408) 988-0279 • www.cel.com

DATA SUBJECT TO CHANGE WITHOUT NOTICE

ADDITIONAL PARAMETERS

Parameters	NESG2021M05
CCB	0.001 pF
CCE	0.18 pF
LB	0.35 nH
LE	0.16 nH
CCBPKG	0.03 pF
CCEPKG	0.001 pF
CBEPKG	0.03 pF
LBPKG	0.9 nH
LCPKG	1.2 nH
LEPKG	0.17 nH

MODEL TEST CONDITIONS

Frequency: 0.1 to 6 GHz
Bias: $V_{CE} = 2$ V, $I_C = 1$ mA to 10 mA
Date: 09/2003

09/02/2003